Supplementary Material S7

Full Limitation Narratives and Per-Limitation Validity Assessments

S7.1 Introduction

This supplement provides a detailed account of the study's methodological limitations and evaluates the threat each poses to the validity of the reported findings. For every limitation, we describe the nature of the concern, its potential impact on the conclusions, and the measures taken to mitigate it.

S7.2 Limitation Narratives

L1: Limited Expert Panel Size and Homogeneity

Narrative
The Delphi panel consisted of six NHS professionals, all from a single national healthcare system. While purposive sampling ensured deep expertise in cybersecurity, clinical safety, and data protection, the relatively small number of participants and the absence of international perspectives may limit the generalisability of the prioritised taxonomy dimensions. A larger, more diverse panel might produce slightly different weightings.

Validity Assessment

- **Threat to External Validity:** Moderate. The findings may not fully reflect the views of experts from other healthcare systems (e.g., US FDA-regulated environments, EU-wide GDPR regimes).
- **Threat to Internal Validity:** Low. Within the defined expert community, consensus was high (Krippendorff's $\alpha = 0.86$), suggesting reliable judgment.
- **Mitigation:**
  - Purposive selection from multiple organisations (acute trusts, community trusts, national bodies) provides breadth within the NHS context.
  - Sensitivity analysis (leave-one-out) showed weight variation ≤ 0.03, indicating robustness to individual opinions.
  - We explicitly limit claims to the NHS setting and call for replication in other healthcare systems.

L2: Single Healthcare System Context (UK NHS)

Narrative
All clinical workflows, regulatory references (NHS Digital DCB0129), and edge-deployment thresholds (latency < 200 ms, memory < 100 MB) are derived from the UK NHS environment. The extent to which these thresholds and the relative importance of compliance vs. accuracy hold in other regulatory frameworks (e.g., HIPAA, FDA SaMD) is unknown.

Validity Assessment

- **Threat to External Validity:** High. Direct transfer of the taxonomy to non-NHS contexts without validation is not warranted.

- **Threat to Construct Validity:** Low to moderate. The constructs (accuracy, compliance, etc.) are defined with reference to NHS-specific regulations, which may not have identical counterparts elsewhere.
- **Mitigation:**
    - The taxonomy is presented as an NHS-specific framework, not a universal one.
    - We discuss potential adaptations needed for other jurisdictions.
    - Future work plans include a multi-national Delphi round to test cross-context robustness.

L3: Static Dataset and Temporal Validity

Narrative
The training and evaluation datasets (IoT-23, CIC-DDoS2019, etc.) represent snapshots of network threats and medical device behaviours at the time of collection. Cyber-attack patterns and device firmware evolve rapidly; the relative performance of the models may degrade over time if re-trained on older data. Additionally, the taxonomy dimensions themselves (e.g., Edge Readiness) might need re-weighting as hardware advances.

Validity Assessment

- **Threat to External Validity:** Moderate. The models tested against historical attacks may not fully capture future zero-day exploits or novel attack vectors.
- **Threat to Internal Validity:** Low. Within the dataset, the models demonstrate consistent convergence and acceptable performance.
- **Mitigation:**
    - We used multiple public datasets to increase diversity.
    - The taxonomy is designed to be revisited periodically; we outline a planned re-evaluation after 24 months.
    - We recommend continuous monitoring and periodic re-training in the main text.

L4: Model Selection and Benchmark Scope

Narrative
We compared four model architectures (GNN, Transformer, Edge-Fusion, HIMS-CDI MLR), but the selection is not exhaustive. Recent developments, such as graph attention networks with dynamic edge features or large language model-based anomaly detectors, were not included. Thus, the reported performance ranking might not generalise to all possible model families.

Validity Assessment

- **Threat to Internal Validity:** Low. All models were trained with identical seeds and early-stopping criteria; the comparisons within the study are fair.
- **Threat to External Validity:** Moderate. The observed ranking may change with newer architectures that have not been tested.
- **Mitigation:**

- We selected representative baselines covering diverse paradigms (spatial, sequential, fusion, linear).
- The code and data are made available to facilitate benchmarking with additional models.
- We explicitly call for future studies to extend the model zoo.

L5: Subjective AHP Weighting

Narrative
Taxonomy dimension weights were derived using the Analytic Hierarchy Process (AHP) from expert pairwise comparisons. Despite high consistency (CR = 0.019), the weights reflect subjective human judgement, which can be influenced by personal experience and organisational priorities. A different expert panel might prioritise dimensions differently.

Validity Assessment

- **Threat to Construct Validity:** Moderate. The AHP method captures perceived importance rather than objective ground truth.
- **Threat to External Validity:** Moderate. The weights may not be optimal for contexts where, for example, regulatory fines are lower.
- **Mitigation:**
    - The Delphi method ensured iterative refinement and consensus.
    - Leave-one-out sensitivity analysis demonstrated stability.
    - We provide the raw matrices and the computation script for transparency; researchers can recompute weights with their own judgements.

L6: Operationalisation of Taxonomy Dimensions

Narrative
Some dimensions (e.g., Explainability, Edge Readiness) were operationalised using simple proxies: post-hoc SHAP values for explainability and binary compliance with DCB0129 thresholds for edge readiness. These operationalisations, while practical, may not capture the full richness of the concepts (e.g., clinician trust, user-experience, comprehensive regulatory alignment).

Validity Assessment

- **Threat to Construct Validity:** Moderate to high. Incomplete construct measurement can lead to misrepresentation of a model's true suitability.
- **Threat to Internal Validity:** Low. All models were evaluated using the same operational definitions, ensuring fair comparisons.
- **Mitigation:**
    - We clearly define each operationalisation and acknowledge its limitations.
    - Where possible, we used established metrics (e.g., F1-score) and referenced standard guidelines.

- Future work will incorporate more nuanced measures (e.g., user studies for explainability, full regulatory mapping for compliance).

S7.3 Summary of Validity Threats

Limitation	External	Internal	Construct
L1 – Panel size & homogeneity	Moderate	Low	–
L2 – NHS-specific context	High	–	Low–Moderate
L3 – Static dataset	Moderate	Low	–
L4 – Model selection	Moderate	Low	–
L5 – Subjective AHP	Moderate	–	Moderate
L6 – Operationalisation	–	Low	Moderate–High

Overall, the study's internal validity is strong; the principal threats concern external generalisability and construct measurement. These are transparently documented, and mitigation strategies have been implemented where feasible.

S7.4 Conclusion

By systematically enumerating the limitations and their impact on validity, this supplement supports a balanced interpretation of the study's findings. The taxonomy and model benchmarks should be viewed as a robust yet context-specific contribution, with clear avenues for future refinement.